



6 Courses

**Modern Robotics, Course 1:
Foundations of Robot
Motion**

**Modern Robotics, Course 2:
Robot Kinematics**

**Modern Robotics, Course 3:
Robot Dynamics**

**Modern Robotics, Course 4:
Robot Motion Planning and
Control**

**Modern Robotics, Course 5:
Robot Manipulation and
Wheeled Mobile Robots**

**Modern Robotics, Course 6:
Capstone Project, Mobile
Manipulation**



Jun 10, 2025

Aung Ko

has successfully completed the online Specialization

Modern Robotics: Mechanics, Planning, and Control

This specialization provides a rigorous treatment of spatial motion and the dynamics of rigid bodies, employing representations from modern screw theory and the product of exponentials formula. Students with a freshman-level engineering background will quickly learn to apply these tools to analysis, planning, and control of robot motion. Students' understanding of the mathematics of robotics will be solidified by writing robotics software. Students will test their software on a free state-of-the-art cross-platform robot simulator, allowing each student to have an authentic robot programming experience with industrial robot manipulators and mobile robots without purchasing expensive robot hardware.

Kevin M. Lynch
Professor of Mechanical
Engineering
Northwestern
University

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